Harsha Vardhan Reddy Vatte
Software Engineer
+1 (203)-928-8089 | Gmail | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Software Engineer with 2+ years of professional experience designing, building, and maintaining scalable full-stack applications and microservices. Proactively embodies a builder mindset by integrating AI coding tools and prompt-to-feature workflows to accelerate development velocity. Proven track record of working cross-functionally with product, design, and data teams to own features end-to-end. Skilled in React Native mobile development, Node.js/Express, Java, and modern testing lifecycles.

TECHNICAL SKILLS

Languages: Java, SQL, JavaScript, TypeScript, Python
Frameworks & Backend: Spring Boot, Node.js, Express, React, React Native, Spring Cloud, Spring Data JPA, Hibernate, Kafka, REST APIs,
Databases: MySQL, MongoDB, PostgreSQL
DevOps & Tools: Docker, Docker Compose, AWS EC2, Cloud Watch, Jenkins, Maven, Git/GitHub/GitLab, CI/CD Pipelines, Postman, JUnit 5, Mockito, Redis, Jest, Cypress
Architecture & AI: Prompt-to-feature workflows, Microservices, Event-Driven Design, Tiny Llama, Ollama LLMs

PROFESSIONAL EXPERIENCE

Optum (UnitedHealth Group) – Java Developer — Healthcare Domain – Apr 2022 – Nov 2023 | Hyderabad, India
Technologies & Tools: Java, React, React Native, Node.js, Express, Spring Boot, Spring Cloud, REST APIs, MySQL, MongoDB, Hibernate, Docker, Jenkins, Kafka, Microservices

- Optimized high-traffic REST APIs for claims status tracking, boosting throughput by 25% under peak loads via structuring complex, highly efficient SQL queries and Java asynchronous execution.
- Collaborated closely with a growing, agile team to re-architect cross-service communication, replacing monolithic dependencies with stateless APIs and deploying services securely on AWS EC2 with CloudWatch monitoring.
- Owned the complete feature testing lifecycle across environments by authoring comprehensive Jest unit tests, managing data mapping, and leading peer code reviews to mentor early-career engineers.
- Developed distributed event-driven components using Kafka to handle high-throughput, asynchronous messaging between downstream microservices, ensuring eventual data consistency across the platform.
- Built and maintained responsive, high-performance features for React Native mobile applications and React web interfaces to stream live data feeds, ensuring smooth data persistence with relational databases.

Rivier University – Teaching Assistant — Computer Science – Jan 2025 – Dec 2025 | Nashua, NH
Technologies & Tools: Core Java, Git, GitHub

- Graded weekly programming assignments and exams, providing written feedback on code correctness, logical structure, and proper formatting.
- Held regular office hours, utilizing strong communication and interpersonal skills to help students troubleshoot compilation errors, fix runtime bugs, and master core computer science concepts.

KEY PROJECTS

AI Knowledge Chat System: Link
Technologies & Tools: React, TypeScript, Spring Boot, FastAPI, Python, Ollama, Tiny Llama, MongoDB, Redis, Docker Compose

- Independently architected a local AI chat platform to optimize prompt-to-feature workflows, leveraging a Spring Boot orchestrator and Fast API to establish rapid prototyping infrastructure.
- Integrated local LLM runtime environments with custom single-turn prompt engineering and response-cleaning algorithms to optimize low-latency output handling.

Microservices Order Processing System: Link
Technologies & Tools: Java, Spring Boot, Spring Cloud Eureka, Spring Data JPA, Docker, Docker Compose, Maven

- Designed a distributed order processing platform composed of 4 independent microservices using a hybrid communication model of synchronous REST APIs and asynchronous event streaming.
- Integrated Spring Cloud Eureka for dynamic service discovery and scaling, utilizing Spring Data JPA for persistent relational database architectures designed for eventual consistency.

EDUCATION

Rivier University — Master of Science, Computer Science | GPA: 3.7 | Sep 2024 – May 2026 | Nashua, NH